R13

Code No: 117EF

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, May/June - 2019
MACHINE LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define learning. [2]
b) What is the need of a target function? [3]
c) List the four steps in rule-post pruning method. [2]
d) Which is the most commonly used ANN learning technique? List its characteristics. [3]
e) Mention two practical difficulties in applying Bayesian methods. [2]
f) Provide an example of Bayesian belief network. [3]
g) Define kernel function. [2]
h) How does kNN handle the curse of dimensionality? [3]
i) Generate two off springs using mutation from 11101001000. [2]
j) State the problem of crowding in genetic algorithm application. [3]

PART-B

(50 Marks)

2. What are the various disciplines influenced machine learning? With suitable example justify your opinion. [10]

OR

3. "Concept learning is viewed as the task of searching through a large space of hypotheses implicitly defined by the hypothesis representation". Support or oppose this statement with relevant discussion. [10]

4. Define entropy and explain how information gain measures the expected reduction in entropy. [10]

OR

5.a) What is the representational power of perceptrons?
b) Differentiate between perceptron rule and delta rule. [5+5]

6.a) What is the significance of jackknife cross-validation test in machine learning?
b) How to prevent over-fitting in machine learning? [5+5]

OR

7. Demonstrate Naïve Bayesian classification of news articles. Consider any five articles of your choice. [10]

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8. What are the properties shared by the instance methods, kNN and locally weighted regression? Make a comparison of these approaches with case based reasoning. [10]

OR

9. Explain how the CADET system employs case based reasoning to assist in the conceptual design of simple mechanical devices such as water faucets. [10]

10.a) State schema theorem utilized in genetic algorithm.
b) How does genetic programming handle block-stacking problem? [5+5]

OR

11.a) What is the essential difference between analytical and inductive learning methods?
b) Describe the explanation based learning algorithm, PROLOG-EBG. [5+5]

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